Attacking a hard olympiad problem

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L5-ST1 Level 5 Advanced · Strategy

How strong solvers work on BMO2, EGMO and IMO papers, from reading what a problem forces and collecting data, through conjectures, extremal examples and switching approaches, to partial write-ups and managing a four and a half hour paper.
By the end you can turn small cases into a conjectured answer or extremal example and then into a proof; decide when to switch approach and keep a record of attempts; plan a long olympiad paper and write up partial progress so that the progress is credited.
Prerequisites: Writing olympiad solutions (L4-ST1), Invariants, monovariants and extremal arguments (L4-CO1), Olympiad Diophantine equations (L4-NT2). **Contests:** BMO Round 2, EGMO, IMO, USAMO. **Time:** about 90 minutes.

1 Lesson

Here is Problem 1 of BMO Round 2 in January 2026.

For positive integers m and n, let $l(m,n)$ be their least common multiple and $h(m,n)$ their highest common factor. Given a prime $p > 3$, let k be the number of ordered pairs (m,n) of positive integers with $l(m,n) + h(m,n) = p^4$. Determine the smallest possible value of k over all primes $p > 3$.

My advice on a problem like this is to prove nothing for the first twenty minutes. Turn the condition into something you can count, work out k for $p = 5, 7, 11$ and guess the answer. The proof comes afterwards and is short, because the numbers have told you what to prove. This unit is about the hours before you have a solution, and about what to write if you never get one.

1.1 The papers

The IMO and EGMO have the same format. There are two papers on consecutive days, each with three problems and lasting four and a half hours, and each problem receives an integer score out of 7 (clauses 5.1 and 7.2 of both sets of General Regulations). At the IMO, contestants may send written questions about the wording to the Jury during the first half hour. BMO Round 2 is one paper of three and a half hours with four problems worth 10 marks each, and the paper states that one or two complete solutions gain far more credit than partial attempts at all four. In 2026 the Distinction threshold, set so that the top quarter of entrants receive Distinction, was 18 out of 40, so two complete BMO2 solutions place you in the top quarter.

1.2 Read the problem before you touch the problem

Decide first what a complete answer must contain. A "find all" problem needs the answer set, the check that every member works and the proof that nothing else does, and the check is the half that students forget. A "least" or "greatest" problem is two problems, an example and a bound, and each half tells you where to look for the other.

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Then ask what each hypothesis is for. In the BMO2 problem the condition $p > 3$ is there because 2 and 3 behave differently from the other primes, so expect 2 and 3 to appear in the solution. If your proof never uses a hypothesis, look for the hole.

1.3 Small cases first

Example 1 (BMO Round 2 2026, Problem 1).
Solve the problem stated at the start of this unit.

Solution.
Reformulate. Write $g = h(m,n)$, $m = gx$ and $n = gy$ with $\gcd(x,y) = 1$. Then $l(m,n) = gxy$, so the equation reads $g(xy+1) = p^4$. Hence $g = p^j$ and $xy = p^{4-j} - 1$, where $j \le 3$ because $xy \ge 1$, and every such (j,x,y) gives a solution. The number of ordered coprime pairs with $xy = N$ is $2^{\omega(N)}$, where $\omega(N)$ is the number of distinct primes dividing N, since each prime power in N goes wholly into x or wholly into y. So

$$k = 2^{\omega(p-1)} + 2^{\omega(p^2-1)} + 2^{\omega(p^3-1)} + 2^{\omega(p^4-1)}.$$

Data. For $p = 5$ the four numbers are 4, 24, $124 = 2^2 \cdot 31$ and $624 = 2^4 \cdot 3 \cdot 13$, so $k = 2+4+4+8 = 18$. For $p = 7$ they are 6, 48, $342 = 2 \cdot 3^2 \cdot 19$ and $2400 = 2^5 \cdot 3 \cdot 5^2$, so $k = 4+4+8+8 = 24$. For $p = 11$ we get $k = 44$. The conjecture is 18.
A dead end. The data suggests that k increases with p, and you could lose an hour on that. The claim is false, since $p = 17$ gives 16, $288 = 2^5 \cdot 3^2$, $4912 = 2^4 \cdot 307$ and $83520 = 2^6 \cdot 3^2 \cdot 5 \cdot 29$, so $k = 2+4+4+16 = 26 < 44$. What matters is how few primes each $p^e - 1$ can have, so we bound each term separately.
Proof. Every $p^e - 1$ is even, so the first term is at least 2. Also $3 \mid p^2 - 1$, because 3 divides one of $p-1, p, p+1$ and does not divide p, so the second term is at least 4. Next, $p^3 - 1 = (p-1)(p^2+p+1)$ and p^2+p+1 is odd and greater than 1, so $\omega(p^3-1) \ge 2$. Finally $p^4 - 1 = (p^2-1)(p^2+1)$. As p is odd and not a multiple of 3, we have $p^2 + 1 \equiv 2 \pmod 4$ and $p^2 + 1 \equiv 2 \pmod 3$, so $\frac{p^2+1}{2}$ is greater than 1 and has a prime factor other than 2 and 3. Hence $\omega(p^4-1) \ge 3$ and

$$k \ge 2+4+4+8 = 18,$$

with equality at $p = 5$. The answer is 18.
Remark. Each of the four bounds came from asking why the corresponding number for $p = 5$ could not have fewer prime factors. So the data gave the answer and also the shape of the proof.

Example 2 (IMO 2017, Problem 1).
For each integer $a_0 > 1$ define $a_{n+1} = \sqrt{a_n}$ if $\sqrt{a_n}$ is an integer, and $a_{n+1} = a_n + 3$ otherwise. Determine all a_0 for which some number A satisfies $a_n = A$ for infinitely many n.

Solution.
Data. Some starting values give

$$3,6,9,3,\ldots \qquad 4,2,5,8,11,\ldots \qquad 7,10,13,16,4,2,\ldots \qquad 12,15,\ldots,36,6,9,3,\ldots$$

The values $3,6,9,12$ end in the cycle $3,6,9$, while $2,4,5,7,8,10$ all reach the progression $2,5,8,\ldots$ and increase for ever. Adding 3 does not change the residue modulo 3, and squares are 0 or 1 modulo 3, so we conjecture that the answer is the multiples of 3.

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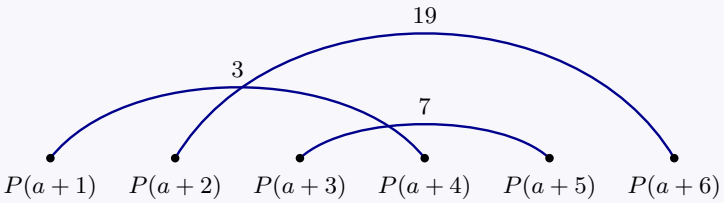
Residue 2. If $a_n \equiv 2 \pmod 3$ then a_n is not a square, so $a_{n+1} = a_n + 3 \equiv 2$. From then on the sequence increases, and no value occurs infinitely often.
Residue 0. Every term is a multiple of 3, since $3 \mid b^2$ implies $3 \mid b$. Working backwards from the goal, we only need the sequence to be bounded, because a bounded sequence of integers takes some value infinitely often. Suppose $a_n > 9$ and choose $m \geq 1$ with $9m^2 < a_n \leq 9(m+1)^2$. Adding 3 repeatedly runs through every multiple of 3 from a_n upwards, so the sequence meets a square no larger than $9(m+1)^2$ and then drops to at most $3(m+1)$, which is less than $9m^2 < a_n$. Hence the terms eventually fall to 9 or below, and $3 \to 6 \to 9 \to 3$ repeats.
Residue 1. No term equals 1, since $a_{n+1} = 1$ forces $a_n = 1$, and $a_0 > 1$. Suppose $a_n \equiv 1 \pmod 3$, so $a_n \geq 4$. The sequence climbs to c^2, where c is the least integer with $c^2 \geq a_n$ and $3 \nmid c$, and then drops to c. One of two consecutive integers is not a multiple of 3, so $c \leq \sqrt{a_n} + 2 < a_n$ when $a_n \geq 5$, and $c = 2$ when $a_n = 4$. If $c \equiv 2$ we are in the first case. Otherwise $c \equiv 1$ and $4 \leq c < a_n$, and we repeat. A decreasing sequence of positive integers is finite, so the residue 2 is eventually reached.
The answer is all multiples of 3.
Remark. Ten starting values fixed the answer and the case split in five minutes. The data said nothing about why residue 1 escapes, which is the hard case, so be honest with yourself about which part of a problem the data has settled.

1.4 The example and the bound together

Example 3 (IMO 2016, Problem 4).
A set of positive integers is fragrant if the set has at least two elements and each element has a prime factor in common with at least one other element. Let $P(n) = n^2 + n + 1$. What is the least positive integer b such that $\{P(a+1), P(a+2), \ldots, P(a+b)\}$ is fragrant for some non-negative integer a?

Solution.
First attempt. Search small a. The values $P(1), \ldots, P(7)$ are $3, 7, 13, 21, 31, 43, 57$, and there is some sharing, as 3 divides $P(1), P(4), P(7)$ and 7 divides $P(2), P(4)$. But no short window works for small a, and a hand search will not reach a in the hundreds. After ten minutes, stop searching and ask which primes two values can share.
Structure. Every $P(n)$ is odd. Suppose an odd prime q divides $P(n)$ and $P(n+d)$, where $1 \leq d \leq 4$. Then q divides $P(n+d) - P(n) = d(2n+d+1)$. If $q \mid 2n+d+1$, then $2n+1 \equiv -d$, and $4P(n) = (2n+1)^2 + 3$ gives $q \mid d^2 + 3$. So q divides d or d^2+3. For $d = 1, 2, 3, 4$ this says that adjacent values are coprime, and values 2, 3 or 4 apart can share only 7, 3 or 19 respectively. Also $3 \mid P(n)$ if and only if $n \equiv 1 \pmod 3$, and $7 \mid P(n)$ if and only if $n \equiv 2$ or $4 \pmod 7$.
Bound. Write $Q_i = P(a+i)$. For $b = 2$ and $b = 3$ some element has only adjacent values available. For $b = 4$, Q_2 can pair only with Q_4 and Q_3 only with Q_1, both through 7, so 7 divides all four values, but only two residues modulo 7 are allowed. For $b = 5$, Q_3 pairs with Q_1 or Q_5 through 7. In the first case $a+1 \equiv 2$ and $a+3 \equiv 4 \pmod 7$, so 7 divides neither Q_2 nor Q_4. Then Q_2 must pair with Q_5 and Q_4 with Q_1, both through 3, so $a+2 \equiv 1$ and $a+4 \equiv 1 \pmod 3$, a contradiction. The second case is the mirror image. Hence $b \geq 6$.

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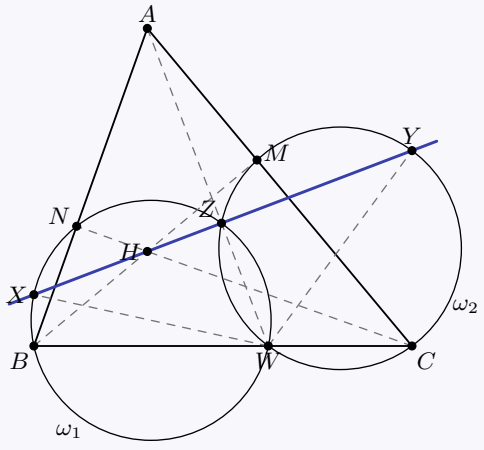
Construction. The bound proof shows what a six-element example must look like, with pairs three apart through 3, two apart through 7 and four apart through 19, as in the figure. Since $P(7) = 57$ and $P(11) = 133 = 7 \cdot 19$, we have $19 \mid P(n)$ when $n \equiv 7$ or $11 \pmod{19}$. So we need $a \equiv 0 \pmod 3$, $a+3 \equiv 2 \pmod 7$ and $a+2 \equiv 7 \pmod{19}$, and the Chinese remainder theorem gives $a = 195$. Then 3 divides $P(196)$ and $P(199)$, 7 divides $P(198)$ and $P(200)$, and 19 divides $P(197)$ and $P(201)$. The answer is $b = 6$.
Remark. The example was read off from the cases where the bound proof could not reach a contradiction. In a "least b" problem, work on the bound and the example in turns. When the bound stalls on a case, build an example from that case.

1.5 In geometry the data is the figure

Draw large, with ruler and compasses, then draw the figure again with the free point moved. A freehand sketch shows what you expect to see, and a fact that survives two accurate figures is worth far more. Look for points that the statement does not mention.

Example 4 (IMO 2013, Problem 4).
Let ABC be an acute triangle with orthocentre H, and let W be a point on the side BC, between B and C. The points M and N are the feet of the altitudes from B and C. Let ω_1 be the circumcircle of BWN and X the point of ω_1 with WX a diameter. Let ω_2 be the circumcircle of CWM and Y the point of ω_2 with WY a diameter. Prove that X, Y and H are collinear.

Solution.
Data. In an accurate figure the circles ω_1 and ω_2 meet again at a point Z, and the line XY passes through Z. Move W and redraw. The point Z moves, but the line WZ still passes through A, and $\angle WZH$ still looks like a right angle. So we bring in Z (a phantom point).



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Proof. Since WX and WY are diameters, $\angle WZX = \angle WZY = 90^\circ$, so X and Y lie on the line through Z perpendicular to WZ. We only need $\angle WZH = 90^\circ$. The points B, C, M, N lie on the circle with diameter BC, so $AN \cdot AB = AM \cdot AC$. These are the powers of A with respect to ω_1 and ω_2, so A lies on the radical axis WZ and $AZ \cdot AW = AN \cdot AB$. Let L be the foot of the altitude from A. The right angles at L and N put B, L, H, N on one circle, so $AH \cdot AL = AN \cdot AB = AZ \cdot AW$. Hence Z, W, L, H lie on one circle. Since $\angle WLH = 90^\circ$, the segment WH is a diameter of that circle, and so $\angle WZH = 90^\circ$. (If $W = L$, the equation gives $AZ = AH$, so $Z = H$ and there is nothing to prove.)
Remark. Without the second figure you are unlikely to guess that A, Z, W are collinear, and once Z is drawn each step is a standard fact. Angle chasing with X and Y directly goes nowhere, since they come from two different circles.

How strong solvers lose problems they could solve.
They try to prove a statement before they know the answer, or trust a pattern from too little data, like the false monotonicity in Example 1. They forget the example in a "least" problem or the check in a "find all" problem. They stay on one approach for two hours, and they leave the write-up to the last ten minutes.

1.6 When to switch, and the written record

If an approach has produced no new true statement in forty minutes (a lemma, a reduction, a solved special case), switch. A long computation with checkable intermediate results is slow, and slow is fine. Reformulations that keep returning to the same equation mean you are stuck. Dead ends often leave a lemma behind, as the abandoned search in Example 3 did, so number your attempts, box each proved claim and write one line on why each failed idea failed. After three hours you will not remember what you have tried, and the record is also where your write-up comes from.

When stuck, do something concrete. Work backwards from the conclusion, as in Example 2, where "some value occurs infinitely often" became "the sequence is bounded". Specialise to $n = 1, 2, 3$ or to an equilateral triangle. Or generalise, since a stronger statement can carry an induction that the original cannot.

1.7 What to write when you have not finished

At the IMO and EGMO each problem is scored from 0 to 7 under a marking scheme that the Jury approves for that problem, so you cannot know in advance which steps carry marks. Make every proved step easy to find. Write the answer and the verified example, and in a "find all" problem the check direction in full. State each proved lemma as a numbered claim with a complete proof, even if you could not use the claim. If the proof has one gap, say where, since "I have not proved Claim 3" is easier to credit than a gap the marker has to find. A check of $n = 1, 2, 3$ alone is rarely worth anything, and your rough record is worth nothing.

1.8 Four and a half hours

Spend the first twenty to thirty minutes trying all three problems and writing down any answer you can guess, then choose an order. Every forty-five minutes, decide honestly whether to continue or switch. Write a solution as soon as you have one, while the details are fresh, because a

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complete write-up often takes thirty minutes, and keep the last forty-five minutes for writing. On BMO2 the same plan gives about fifty minutes per problem. Late in a long paper, go back to fresh small cases rather than repeat a failed attempt.

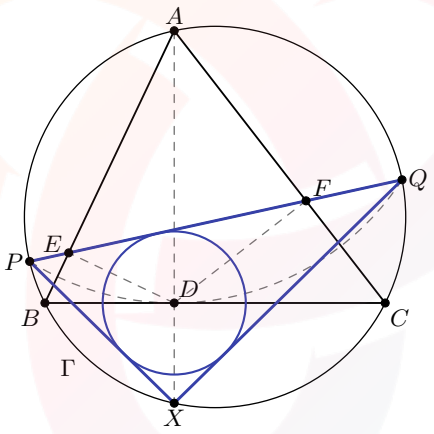
2 Practice problems

The problems are roughly in order of difficulty, from ⋆⋆⋆ to ⋆⋆⋆⋆. Hints are in Section 4 and full solutions in Section 5.

Problem 1 ⋆⋆⋆ Find all pairs (n, k) of positive integers such that $2^n + n^k$ divides $4^n + n^{2k}$.

Process. Find the answer from data, then prove the answer.

Problem 2 ⋆⋆⋆⋆ Let ABC be an acute triangle with circumcircle Γ, and let D be the foot of the altitude from A. Let E and F be the feet of the perpendiculars from D to AB and AC. The line EF meets Γ at P and Q, and the line AD meets Γ again at X. Prove that D is the incentre of triangle PQX.



The rough work of a student reads *"Invert about A with radius AD. This swaps B with E and C with F, so Γ goes to line EF and P, Q are fixed. Hence $AP = AQ = AD$, and the incentre and excentre lemma finishes."*

Process. Turn a sketch into a complete solution.

Problem 3 ⋆⋆⋆⋆ Let $n \geq 6$, and let $x_1, x_2, \ldots, x_n$ be non-negative real numbers with $x_1 + x_2 + \cdots + x_n = 1$. Find the largest possible value of

$$x_1x_2x_3 + x_2x_3x_4 + \cdots + x_{n-1}x_nx_1 + x_nx_1x_2.$$

A student writes *"Let A, B and C be the sums of the x_i with $i \equiv 1, 2$ and 0 modulo 3. Every term is a product of one variable from each class, so the sum is at most $ABC \leq \frac{1}{27}$."*

Process. Rescue a failing approach.

Problem 4 ⋆⋆⋆⋆ Let $n \geq 3$. Find the largest number of 3-element subsets of $\{1, 2, \ldots, n\}$ that can be chosen so that any two of the chosen subsets have either 0 or 2 elements in common.

Process. Find the extremal configuration from small cases, then prove a matching bound.

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Problem 5 ⋆ ⋆ ⋆⋆ For a non-negative integer n, let $f(n)$ be the number of sequences $(a_0, a_1, a_2, \dots)$ with every $a_i \in \{0, 1, 2, 3\}$, only finitely many a_i non-zero, and

$$n = a_0 + 3a_1 + 9a_2 + 27a_3 + \cdots .$$

For each positive integer m, find the largest value of $f(n)$ over $0 \le n < 3^m$.

Process. Find the answer from data, then prove the answer.

Problem 6 ⋆ ⋆ ⋆⋆ The numbers $1, 2, \dots, n$ are written on a board, where $n \ge 2$. A move consists of erasing two numbers a and b and writing $\frac{a+b}{2}$. After $n-1$ moves a single number remains. Find the largest possible value of this number.

Process. Find the extremal strategy from small cases, then prove that no strategy does better.

3 Contest ladder

Real past problems on the ideas of this unit, easiest first. Each source is a link to the official paper or the AoPS wiki; the note under it says where to start, not how to finish.

1. **EGMO 2023 Problem 1** ⋆ ⋆ ⋆
   Look at the largest and the smallest of the numbers
2. **USAMO 2014 Problem 1** ⋆ ⋆ ⋆
   Guess the extremal polynomial first, then find an expression for the product in terms of P
3. **UKMT BMO2 2025 Problem 1** ⋆ ⋆ ⋆
   Write down every sum of consecutive reciprocals you can evaluate, then ask what is missing
4. **UKMT BMO2 2024 Problem 2** ⋆ ⋆ ⋆
   Find the linear solutions, then study the sequence n, f(n), f(f(n)), and so on
5. **IMO 2011 Problem 1** ⋆ ⋆ ⋆
   Decide first which pairs can never work, then collect data on the rest
6. **IMO 2015 Problem 1** ⋆ ⋆ ⋆
   Build examples for n = 3, 4, 5 before deciding which n can be centre-free
7. **UKMT BMO2 2025 Problem 3** ⋆ ⋆ ⋆⋆
   Find the answer for every n up to 5 before proving anything
8. **UKMT BMO2 2024 Problem 4** ⋆ ⋆ ⋆⋆
   Tabulate small pairs (m, n), conjecture the answer, then look for an invariant
9. **IMO 2014 Problem 2** ⋆ ⋆ ⋆⋆
   Find the answer for small n, then build the configuration that is worst for the solver
10. **EGMO 2023 Problem 3** ⋆ ⋆ ⋆⋆
    Small k first, then find a dictionary that shows your threshold is sharp

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4 Hints

1. Compare $4^n + n^{2k}$ with $(2^n + n^k)^2$. Then write $n = 2^v w$ with w odd, and compare the power of 2 dividing $2^n + n^k$ with the power of 2 dividing n^k.

2. The outline is right. A complete solution must prove $AE \cdot AB = AD^2$ and $AP = AQ = AD$ without taking the inversion on trust, and must decide whether D is the incentre or an excentre. For the last point, find on which side of the line PQ the point X lies.

3. Test the student's claim for $n = 7$ on the terms that wrap round the cycle. The argument is correct when $3 \mid n$. For the other values of n, find a way to remove one variable without decreasing the sum, and find where $n \geq 6$ is needed.

4. The first guess, all triples containing 1 and 2, gives $n - 2$. Compare with what you can do for $n = 8$. For the bound, show that if two chosen triples share a pair $\{a, b\}$, then every chosen triple meeting their union either contains $\{a, b\}$ or lies inside their union.

5. Show that $f(3k+1) = f(3k+2) = f(k)$ and $f(3k) = f(k) + f(k-1)$ for $k \geq 1$. The maxima for $m = 1, \ldots, 5$ form a famous sequence. For the proof, induct on m with a second statement about $f(k) + f(k-1)$.

6. For $n = 3$ and $n = 4$ compare all strategies. Show that the final number is $\sum 2^{-d_i} i$ for positive integers d_i with $\sum 2^{-d_i} = 1$. Then bound the sum of the k largest weights by counting binary digits.

5 Solutions

Problem 1

Find all pairs (n, k) of positive integers such that $2^n + n^k$ divides $4^n + n^{2k}$.

Process. Find the answer from data, then prove the answer.

Data. A search with $n \leq 300$ and $k \leq 11$ finds only $(2, 2)$, $(4, 2)$ and $(16, 4)$. In each of these n is a power of 2, and n^k equals 4, 16 and 65536, which are 2^n in each case. We conjecture that the solutions are the pairs with n a power of 2 and $n^k = 2^n$. The next one is $(256, 32)$, far outside the search, so the data alone could never have settled the question.

Reduction. Let $d = 2^n + n^k$. Since

$$4^n + n^{2k} = (2^n + n^k)^2 - 2^{n+1} n^k,$$

the condition is equivalent to

$$d \mid 2^{n+1} n^k. \qquad (1)$$

Write $n = 2^v w$ with w odd and $v \geq 0$. Then $2^{n+1} n^k = 2^{n+1+kv} w^k$, so by (1) the odd part of d divides w^k and is at most w^k.

Case $n \neq kv$. The highest powers of 2 dividing 2^n and $n^k = 2^{kv} w^k$ are 2^n and 2^{kv}, which are different, so the highest power of 2 dividing d is $2^{\min(n, kv)} \leq 2^{kv}$. Hence $d \leq 2^{kv} w^k = n^k < d$, a contradiction.

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Case $n = kv$. Here $v \geq 1$ and $d = 2^{kv} + 2^{kv}w^k = 2^{kv}(1 + w^k)$, so (1) becomes $1 + w^k \mid 2^{kv+1}w^k$. As $1 + w^k$ is coprime to w, the number $1 + w^k$ divides 2^{kv+1} and is a power of 2. Suppose $w \geq 3$. If k is even then $w^k \equiv 1 \pmod 8$, so $1 + w^k \equiv 2 \pmod 8$ and $1 + w^k = 2$, which is false. If $k \geq 3$ is odd then $1 + w^k = (1 + w) \cdot \frac{1+w^k}{1+w}$, and the second factor equals $w^{k-1} - w^{k-2} + \cdots - w + 1$, a sum of k odd numbers with signs, so the second factor is odd. The second factor is also greater than 1, since $1 + w^k > 1 + w$, so $1 + w^k$ is not a power of 2. If $k = 1$ then $n = v$, but $n = 2^v w > v$. So $w = 1$ and $n = 2^v = kv$. Then v divides 2^v, so $v = 2^s$ for some $s \geq 0$, and $n = 2^{2^s}$, $k = \frac{2^v}{v} = 2^{2^s - s}$.

These work. For $n = 2^{2^s}$ and $k = 2^{2^s - s}$ we have $n^k = 2^{2^s k} = 2^{2^{2^s}} = 2^n$. So $d = 2^{n+1}$ and $4^n + n^{2k} = 2 \cdot 4^n = 2^{2n+1}$, which is divisible by 2^{n+1}.

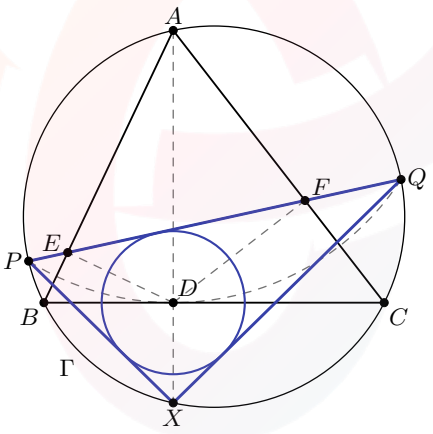
Conclusion. The pairs are $(n, k) = \left(2^{2^s}, 2^{2^s - s}\right)$ with $s \geq 0$.

Remark. The data pointed at $n^k = 2^n$, and the proof explains the pattern. Unless the two terms of $2^n + n^k$ carry the same power of 2, the odd part of d is too small to make up the size of d.

Answer: $(n, k) = \left(2^{2^s}, 2^{2^s - s}\right)$ for $s \geq 0$, that is, $(2, 2), (4, 2), (16, 4), (256, 32), \ldots$

Problem 2

Let ABC be an acute triangle with circumcircle Γ, and let D be the foot of the altitude from A. Let E and F be the feet of the perpendiculars from D to AB and AC. The line EF meets Γ at P and Q, and the line AD meets Γ again at X. Prove that D is the incentre of triangle PQX.



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Process. Turn a sketch into a complete solution.

Let O be the centre of Γ.

Step 1. $AE \cdot AB = AF \cdot AC = AD^2$. Triangle ABD has a right angle at D, so the angles at A and B are acute and E is an interior point of the segment AB. Triangles AED and ADB have right angles at E and D and share the angle at A, so they are similar and $AE \cdot AB = AD^2$. In the same way F is an interior point of AC and $AF \cdot AC = AD^2$.

Step 2. $AP = AQ$. By Step 1, $AE \cdot AB = AF \cdot AC$ with E and F on the segments AB and AC, so B, C, F, E are concyclic and $\angle AEF = \angle ACB$. Let T be a point on the tangent to Γ at A, on the opposite side of AB from C. The tangent and chord angle gives $\angle TAB = \angle ACB = \angle AEF$. These are alternate angles for the lines AT and EF cut by AB, as T and F lie on opposite sides of AB, so $AT \parallel EF$. Since $OA \perp AT$, the line OA is perpendicular to the chord PQ, so OA is the perpendicular bisector of PQ and $AP = AQ$.

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Step 3. $AP = AQ = AD$**.** The point E lies inside Γ and on the line PQ, so E is on the chord PQ. As E is an interior point of the segment AB, the points A and B lie on opposite sides of PQ, and B lies on the arc PQ not containing A. Label P and Q so that A, P, B, Q lie on Γ in this order. Then B and Q are on the same arc cut off by the chord AP, so

$$\angle ABP = \angle AQP = \angle APQ = \angle APE,$$

using $AP = AQ$ and the fact that E lies on the segment PQ. Triangles APE and ABP also share the angle at A, so they are similar and

$$\frac{AP}{AB} = \frac{AE}{AP}, \qquad AP^2 = AE \cdot AB = AD^2.$$

Hence $AP = AQ = AD$.

Step 4. X **lies on the arc** PQ **not containing** A**, and** D **lies between** A **and** X**.** The triangle is acute, so D is an interior point of the segment BC and lies inside Γ. Hence D is between A and X on the chord AX. Since E and F are interior points of AB and AC, the points B and C, and so the whole segment BC, lie on the opposite side of the line EF from A. So D is on that side, and so is X, because the ray from A through D crosses the line EF once, before reaching D. Thus X is on the arc PQ not containing A, and X is neither P nor Q.

Step 5. The incentre of PQX **is** D**.** Let J be the incentre of triangle PQX, and write $\angle PXQ = 2x$ and $\angle QPX = 2y$. By Steps 2 and 4, A is the midpoint of the arc PQ not containing X. So XA bisects the angle PXQ, and J lies on the segment XA. The order of the points on Γ is A, P, X, Q, so P and X are on the same side of the chord AQ and $\angle APQ = \angle AXQ = x$. As A and J lie on opposite sides of PQ,

$$\angle APJ = \angle APQ + \angle QPJ = x + y.$$

The exterior angle at J of triangle XJP is $\angle AJP = \angle JXP + \angle JPX = x + y$. So triangle AJP is isosceles with $AJ = AP$. Now J and D are both points of the segment AX at distance AP from A, so $J = D$.

What the sketch left out. The sketch assumed the inversion facts that Steps 1 to 3 prove, and never decided between the incentre and the excentre opposite X, which also lies on the line AX at distance AP from A. Step 4 is what settles that, and a marker would expect to see the step.

Problem 3

Let $n \geq 6$, and let $x_1, x_2, \ldots, x_n$ be non-negative real numbers with $x_1 + x_2 + \cdots + x_n = 1$. Find the largest possible value of

$$x_1x_2x_3 + x_2x_3x_4 + \cdots + x_{n-1}x_nx_1 + x_nx_1x_2.$$

A student writes "Let A, B and C be the sums of the x_i with $i \equiv 1, 2$ and 0 modulo 3. Every term is a product of one variable from each class, so the sum is at most $ABC \leq \frac{1}{27}$."

Process. Rescue a failing approach.

Where the attempt fails. When $3 \mid n$ the classes go round the cycle consistently. Every term $x_ix_{i+1}x_{i+2}$ then takes one variable from each class, different terms are different products in the expansion of ABC, and AM-GM gives

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$$x_1x_2x_3 + \cdots + x_nx_1x_2 \le ABC \le \left(\frac{A+B+C}{3}\right)^3 = \frac{1}{27}.$$

When $3 \nmid n$ the classes break where the cycle closes. For $n = 7$ the term $x_7x_1x_2$ contains x_7 and x_1, whose indices are both 1 modulo 3, so this term is not one of the products in ABC and the inequality sum $\le ABC$ can fail. The bound $\frac{1}{27}$ may still hold, and numerical search for $n = 7$ and $n = 8$ finds nothing above $\frac{1}{27}$. The hypothesis $n \ge 6$ matters, since equal variables give $\frac{4}{64} = \frac{1}{16}$ for $n = 4$ and $\frac{5}{125} = \frac{1}{25}$ for $n = 5$, both larger than $\frac{1}{27}$.

The value is attained at $x_1 = x_2 = x_3 = \frac{1}{3}$ with all other variables 0.

Removing a variable. Let $n \ge 7$ and take indices modulo n. Choose j with x_j smallest. Delete x_j and replace x_{j+1} by $y = x_j + x_{j+1}$, keeping the cyclic order, so we get $n - 1$ variables with sum 1. The indices $j-2, j-1, \ldots, j+3$ are six different indices, and the only terms that change are the four old terms

$$x_{j-2}x_{j-1}x_j + x_{j-1}x_jx_{j+1} + x_jx_{j+1}x_{j+2} + x_{j+1}x_{j+2}x_{j+3},$$

which are replaced by the three new terms

$$x_{j-2}x_{j-1}\,y + x_{j-1}\,y\,x_{j+2} + y\,x_{j+2}x_{j+3}.$$

Expanding, the new terms minus the old terms equal

$$x_{j-1}x_{j+1}(x_{j-2} - x_j) + x_{j+1}x_{j+2}(x_{j-1} - x_j) + x_{j-1}x_jx_{j+2} + x_jx_{j+2}x_{j+3},$$

which is non-negative because x_j is the smallest variable. So the sum does not decrease.

The bound. Starting from any $n \ge 7$, repeat the step until $n = 6$. The sum has not decreased, and for $n = 6$ the grouping argument applies since $3 \mid 6$. Hence the sum is at most $\frac{1}{27}$ for every $n \ge 6$.

Conclusion. The largest value is $\frac{1}{27}$.

Remark. The grouping was right, but only when $3 \mid n$. The rescue reduces every other case to $n = 6$ instead of hunting for a new grouping, and the hypothesis $n \ge 6$ is what the step needs, because the six indices $j-2, \ldots, j+3$ must be different.

Answer: $\frac{1}{27}$

Problem 4

Let $n \ge 3$. Find the largest number of 3-element subsets of $\{1, 2, \ldots, n\}$ that can be chosen so that any two of the chosen subsets have either 0 or 2 elements in common.

Process. Find the extremal configuration from small cases, then prove a matching bound.

Data. The first guess is a star, all triples containing 1 and 2, with $n - 2$ triples. But all four triples of $\{1, 2, 3, 4\}$ also work, since any two of them share two elements, and two such blocks on $\{1, \ldots, 8\}$ give $8 > 6$. For $n = 3, 4, \ldots, 10$ the maxima are $1, 4, 4, 4, 5, 8, 8, 8$, which fit $\max(4\lfloor n/4 \rfloor, n - 2)$.

Constructions. Disjoint blocks of four elements, with all four triples of each block, give $4\lfloor n/4 \rfloor$ triples. A star gives $n - 2$.

Lemma 1. Let $A = \{a, b, c\}$ and $B = \{a, b, d\}$ be chosen, and let C be another chosen triple that meets $A \cup B$. Then C contains $\{a, b\}$ or $C \subseteq \{a, b, c, d\}$.

Proof. Suppose C does not contain both a and b. If C contains exactly one of them, say a, then $C \cap A$ and $C \cap B$ are non-empty, so both have two elements, which forces $c \in C$ and $d \in C$, and

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$C = \{a, c, d\}$. If C contains neither, then $C \cap A \subseteq \{c\}$ has 0 elements and similarly $d \notin C$, so C misses $A \cup B$, against the assumption.

Components. Join two chosen triples when they share two elements, and split the chosen triples into connected components. Triples in different components share no element, so different components use disjoint sets of elements.

Lemma 2. In each component, either all triples contain a common pair (call the component a star), or all triples lie in one 4-element set and there are at least three of them (call the component a block).

Proof. A component with one triple, or with two triples, is a star. Otherwise take adjacent triples $A = \{a, b, c\}$ and $B = \{a, b, d\}$ in the component K and let $Q = \{a, b, c, d\}$. Suppose first that K has a third triple containing $\{a, b\}$. If some triple of K does not contain $\{a, b\}$, then by connectivity some such triple T is adjacent to a triple $T' = \{a, b, t\}$ of K. Since $T \cap T'$ has two elements and T does not contain $\{a, b\}$, we have $t \in T$, and T contains a or b. Among the three triples of K through $\{a, b\}$ pick two, $\{a, b, x\}$ and $\{a, b, y\}$, other than T'. By Lemma 1 for this pair, $T \subseteq \{a, b, x, y\}$, which is false because $t \notin \{a, b, x, y\}$. So K is a star. Suppose instead that A and B are the only triples of K containing $\{a, b\}$. A triple of K adjacent to a triple inside Q meets $Q = A \cup B$, so by Lemma 1 the triple is A, B or a subset of Q. Starting from A and moving along adjacent triples, every triple of K lies inside Q. With at least three triples, K is a block.

Bound. A star with k triples uses $k + 2$ elements, so a star on v elements has $v - 2$ triples. A block uses 4 elements and has at most 4 triples. Suppose there are b blocks and s stars on $v_1, \ldots, v_s$ elements, and u elements are unused, so $n = 4b + v_1 + \cdots + v_s + u$. The number of triples is at most

$$4b + (v_1 - 2) + \cdots + (v_s - 2) = n - u - 2s.$$

If $s = 0$ this is at most $4b \leq 4\lfloor n/4 \rfloor$, and if $s \geq 1$ this is at most $n - 2$.

Conclusion. The maximum is $\max(4\lfloor n/4 \rfloor, n - 2)$, which is n, $n - 1$, $n - 2$ and $n - 2$ for $n \equiv 0, 1, 2, 3 \pmod 4$.

Remark. The first guess, a star, is the worst kind of component, since a star wastes two elements. The block of four elements wastes nothing, and the answer depends on n modulo 4 because blocks only fit in groups of four.

Answer: $\max\big(4\lfloor n/4 \rfloor,\ n - 2\big)$, that is, n, $n - 1$, $n - 2$, $n - 2$ for $n \equiv 0, 1, 2, 3 \pmod 4$

Problem 5

For a non-negative integer n, let $f(n)$ be the number of sequences $(a_0, a_1, a_2, \ldots)$ with every $a_i \in \{0, 1, 2, 3\}$, only finitely many a_i non-zero, and

$$n = a_0 + 3a_1 + 9a_2 + 27a_3 + \cdots.$$

For each positive integer m, find the largest value of $f(n)$ over $0 \leq n < 3^m$.

Process. Find the answer from data, then prove the answer.

Recurrence. In a representation of n, the digit a_0 is congruent to n modulo 3, and the remaining digits represent $\frac{n - a_0}{3}$. If $n = 3k + 1$ or $n = 3k + 2$ then a_0 is 1 or 2 respectively, so $f(3k + 1) = f(3k + 2) = f(k)$. If $n = 3k$ with $k \geq 1$ then a_0 is 0 or 3, so $f(3k) = f(k) + f(k - 1)$. Also $f(0) = 1$.

Data. The values $f(0), \ldots, f(26)$ are $1, 1, 1, 2, 1, 1, 2, 1, 1, 3, 2, 2, 3, 1, 1, 2, 1, 1, 3, 2, 2, 3, 1, 1, 2, 1, 1$, and the maxima for $m = 1, 2, \ldots, 6$ are $1, 2, 3, 5, 8, 13$. We conjecture that the answer is F_{m+1}.

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Bound. We prove by induction on $m \geq 0$ the two statements

$$(\mathrm{A}_m)\ \ f(n) \leq F_{m+1} \text{ for } 0 \leq n < 3^m, \qquad (\mathrm{B}_m)\ \ f(k) + f(k-1) \leq F_{m+2} \text{ for } 1 \leq k < 3^m.$$

For $m = 0$, (A_0) says $f(0) \leq 1$ and (B_0) is empty. Let $m \geq 1$ and assume (A_{m-1}) and (B_{m-1}). Write $n = 3j + r$ or $k = 3j + r$ with $r \in \{0, 1, 2\}$, so that $0 \leq j < 3^{m-1}$.

For (A_m), if $r \neq 0$ then $f(n) = f(j) \leq F_m \leq F_{m+1}$. If $r = 0$ and $j = 0$ then $f(n) = 1$. If $r = 0$ and $j \geq 1$ then $f(n) = f(j) + f(j-1) \leq F_{m+1}$ by (B_{m-1}).

For (B_m), if $r = 2$ then $f(k) + f(k-1) = f(j) + f(j) \leq 2F_m \leq F_{m+2}$. If $r = 1$ and $j = 0$ then $k = 1$ and $f(1) + f(0) = 2 \leq F_{m+2}$. If $r = 1$ and $j \geq 1$ then

$$f(k) + f(k-1) = f(j) + \big(f(j) + f(j-1)\big) \leq F_m + F_{m+1} = F_{m+2}.$$

If $r = 0$ then $j \geq 1$ and $k - 1 = 3(j-1) + 2$, so

$$f(k) + f(k-1) = \big(f(j) + f(j-1)\big) + f(j-1) \leq F_{m+1} + F_m = F_{m+2}.$$

This completes the induction, and (A_m) is the bound.

Construction. For $x \geq 1$ write $P(x) = \big(f(x-1), f(x)\big)$. The recurrence gives

$$P(3x) = \big(f(x-1),\ f(x) + f(x-1)\big), \qquad P(3x+1) = \big(f(x) + f(x-1),\ f(x)\big).$$

Let $x_1 = 1$, so $P(x_1) = (1, 1)$. Given x_t, let $x_{t+1} = 3x_t$ if $f(x_t - 1) \geq f(x_t)$, and $x_{t+1} = 3x_t + 1$ otherwise. Each step keeps the larger entry of P and replaces the smaller entry by the sum of the two, so by induction the entries of $P(x_t)$ are F_t and F_{t+1}. Also $x_t < 3^t$, since $x_1 = 1 < 3$ and $x_{t+1} \leq 3x_t + 1 \leq 3(3^t - 1) + 1 < 3^{t+1}$. So one of $x_m - 1$ and x_m is a number below 3^m at which f equals F_{m+1}.

Conclusion. The largest value of $f(n)$ over $0 \leq n < 3^m$ is the Fibonacci number F_{m+1}.

Remark. A bound on f alone does not survive the recurrence, because $f(3k)$ involves two consecutive values. Statement (B_m) about consecutive pairs is the stronger hypothesis that makes the induction close, and the same pairs drive the construction.

Answer: F_{m+1}, where $F_1 = F_2 = 1$ and $F_{j+2} = F_{j+1} + F_j$ are the Fibonacci numbers

Problem 6

The numbers $1, 2, \ldots, n$ are written on a board, where $n \geq 2$. A move consists of erasing two numbers a and b and writing $\frac{a+b}{2}$. After $n - 1$ moves a single number remains. Find the largest possible value of this number.

Process. Find the extremal strategy from small cases, then prove that no strategy does better.

Data. For $n = 3$ the three strategies end with $\frac{9}{4}$, 2 and $\frac{7}{4}$, and the best one merges 1 and 2 first. For $n = 4$ the best result is $\frac{25}{8}$, again from merging the current result each time with the smallest number left. We conjecture that this greedy strategy is optimal. The greedy strategy ends with

$$\frac{n}{2} + \frac{n-1}{4} + \cdots + \frac{2}{2^{n-1}} + \frac{1}{2^{n-1}} = n - 1 + \frac{1}{2^{n-1}}.$$

Weights. Every number on the board is a combination $\sum c_i\, i$ of the original numbers, and a move halves all the coefficients of the two numbers erased. So the final number is $\sum_{i=1}^{n} w_i\, i$ with $w_i =$

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2^{-d_i}, where $d_i \geq 1$ counts the moves that erase a number in which i has a non-zero coefficient. The same moves applied to n copies of 1 leave 1, since the average of two 1s is 1, so $w_1 + \cdots + w_n = 1$.

Lemma. For $1 \leq k \leq n-1$, the sum of any k of the weights is at most $1 - 2^{-k}$.

Proof. Let s be the sum of k of the weights. The other $n - k \geq 1$ weights are positive, so $s < 1$. Each weight is a power of $\frac{1}{2}$, and the binary expansion of a sum of two numbers has no more digits 1 than the two expansions have together, so s has at most k digits 1 in binary. A binary fraction below 1 with at most k digits 1 is at most $0.11\ldots1$ with k ones, which is $1 - 2^{-k}$.

Bound. Let W_k be the sum of the k largest weights, with $W_0 = 0$, so $W_k \leq 1 - 2^{-k}$ for $k \leq n-1$ and $W_n = 1$. By the rearrangement inequality the final number is at most the value obtained when the largest weight goes to n, the next largest to $n-1$, and so on. Summing by parts, that value is

$$\sum_{j=1}^{n} (n+1-j)(W_j - W_{j-1}) = W_1 + W_2 + \cdots + W_n.$$

Hence the final number is at most

$$\sum_{k=1}^{n-1} \left(1 - \frac{1}{2^k}\right) + 1 = n - 1 + \frac{1}{2^{n-1}}.$$

Conclusion. The greedy strategy attains $n - 1 + \frac{1}{2^{n-1}}$ and no strategy does better, so this is the largest possible value.

Remark. The greedy strategy gives the weights $\frac{1}{2}, \frac{1}{4}, \ldots, \frac{1}{2^{n-1}}, \frac{1}{2^{n-1}}$, whose partial sums meet the lemma with equality. The step to remember is comparing partial sums of weights rather than individual weights, and the binary digit count is what makes the partial sums easy to bound.

Answer: $n - 1 + \frac{1}{2^{n-1}}$

The online version, with progress tracking, is at kevinvariant08.github.io/pathways.html.

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